Mohammed Shakib Hossain

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Research Interests

Usable Security and Privacy, Human-Computer Interaction (HCI), Human-Centered Security and Privacy, Human Factors in Security, Empirical and Behavioral Security, Authentication and Security Protocols, Security and Privacy in Emerging Technologies

Education

BRAC University Dhaka, Bangladesh

Master of Science in Computer Science and Engineering *Ongoing*

- **Current CGPA: 4.00/4.00**
  Academic Highlights *Courses(Completed/enrolled)*: Introduction to Cryptography, Applied Cryptography, HCI and Design, Information Assurance and Human Factor, Usable Security and Privacy, Computer Communications and Networks, Blockchain and Distributed Ledger, Research Methodology and Seminar

BRAC University Dhaka, Bangladesh

Bachelor of Science in Computer Science and Engineering *June 2021 – Jan 2025*

- **CGPA: 3.93/4.00** ***Highest Distinction***
  Academic Highlights: Consistently earned A/A+ grades in core CSE and mathematics courses, including Programming Languages, Data Structures & Algorithms, Operating Systems, Computer Architecture, Theory of Computation, Compiler Design, Blockchain & Cryptocurrency, Artificial Intelligence, and Advanced Mathematics.

Research Experience

MSc Thesis Research 2026 – Present

Department of Computer Science and Engineering, BRAC University *Dhaka, Bangladesh*

- Investigating security policy violations in organizational workflows, with a focus on how security expectations interact with employees' everyday work practices.
- Conducted preliminary exploratory interviews with software/IT employees to understand their experiences with information sharing, client communication, and security-related practices.
- Reviewed literature on security policy compliance, workplace security practices, and usable security, and developed the research background and related work for the ongoing study.

BSc Thesis Research 2024 – 2025

Department of Computer Science and Engineering, BRAC University *Dhaka, Bangladesh*

- Designed a novel NFT-based authentication mechanism for federated metaverses, addressing usability and privacy limitations of traditional password-based and centralized authentication.
- Conducted a user study to evaluate the usability of the proposed authentication mechanism using the System Usability Scale and a custom UX survey grounded in ISO 9241-11 and Nielsen's heuristics and analyzed the collected user feedback.
- Developed a proof-of-concept implementation and integrated the proposed mechanism with a VR-based metaverse environment.
- Implemented the system on a Minecraft Paper server, where gameplay access was gated by successful NFT-based wallet authentication, with a Next.js/Node.js frontend and MetaMask wallet integration.
- Developed a threat model and formally verified the proposed authentication protocols using ProVerif.
- Repository: [BRAC University Repository]

Publications

Conference Papers

- **M. S. Hossain**, K. E. U. Hawak, M. Jawad, M. Y. Ali, and M. S. Ferdous, *"A Secure Authentication Mechanism for Federated Metaverses Using Non-Fungible Tokens (NFTs),"* accepted for publication at the *IEEE International Conference on Blockchain (IEEE Blockchain)*, 2026. [Conference]

Manuscripts in Preparation

- **M. S. Hossain**, *"Understanding Security Policy Violations and Exceptions in Organizational Workflow in Bangladesh,"* manuscript in preparation, 2026.

Research Methods

Qualitative Methods: Semi-structured Interviews, Reflexive Thematic Analysis (Braun & Clarke)
User Research & Evaluation: User Studies, Usability Evaluation, System Usability Scale (SUS), Likert-scale UX Surveys
Quantitative Methods: Descriptive Statistics, Likert-scale Analysis

Work Experience

Undergraduate Teaching Assistant (Student Tutor) Oct 2023 – May 2025
Department of Computer Science and Engineering, BRAC University *Dhaka, Bangladesh*
Courses taught: Algorithms (CSE221), Discrete Mathematics (CSE230)

- Conducted tutorials and individual consultations, supporting 100+ students in Discrete Mathematics, Algorithms, and problem-solving.
- Prepared problem sets and assessment rubrics aligned with course learning outcomes to reinforce conceptual understanding.
- Evaluated assignments and quizzes and provided feedback to support students learning and performance.
- Collaborated with course instructors to facilitate tutorials and hands-on activities and provide technical guidance to students.

Honors & Awards

100% Merit-Based Graduate Scholarship Dhaka, Bangladesh
Full tuition waiver for outstanding academic merit and performance in the MSc program. *2026 – Present*

100% BRAC Scholarship Dhaka, Bangladesh
Awarded full tuition support for outstanding academic performance in undergraduate studies. *June 2021 – Jan 2025*

Vice Chancellor's List (9 of 11 Semesters) Dhaka, Bangladesh
Recognized for achieving a GPA between 3.90 and 4.00 in individual semesters. *June 2021 – Jan 2025*

Graduated with Highest Distinction Dhaka, Bangladesh
Conferred upon completion of the BSc with a CGPA of 3.80 or above. *Jan 2025*

Projects

Chat Secretly – Cryptography-Based Secure Messaging System | *FastAPI, Python, SQLite* 2026

- Collaborated in a team to develop a cryptography-based secure messaging platform.
- Implemented and analyzed AES-128, DES, RSA, ECC, substitution cipher, and double transposition cipher.
- Developed a Crypto Lab supporting encryption/decryption, key generation, and cryptographic attack demonstrations.
- Integrated user authentication and secure messaging using FastAPI and SQLite.
- Project source: [GitHub Link]

StreamPay – Decentralized Payroll Streaming Protocol | *Solidity, Foundry, Ethers.js v6* 2026

- Built a decentralized payroll streaming protocol enabling companies to lock ETH for employees, unlocking funds continuously (per-second) until fully withdrawn.
- Developed and tested smart contracts in Solidity using the Foundry framework.
- Built a frontend using HTML/CSS/JavaScript and Ethers.js v6 to interact with the deployed contracts.
- Project source: [GitHub Link]

Medical Image Analysis with Explainable AI (XAI) | *PyTorch, CNN, ResNet-50* 2026

- Developed a deep learning pipeline for chest X-ray classification (Normal vs Pneumonia) using a fine-tuned ResNet-50 model.
- Integrated Explainable AI techniques (Grad-CAM, Grad-CAM++) to visualize model decision regions and improve interpretability.
- Implemented data preprocessing, training, evaluation, and visualization for clinically meaningful insights.
- Project source: [GitHub Link]

Transformer-based PDF Summarization System | *BERT, BART, NLP* 2026

- Developed a document summarization pipeline with PDF text extraction, preprocessing, and transformer-based summarization (BART).
- Implemented chunking (map-reduce strategy) to handle long documents and improve summary coherence.
- Evaluated summarization quality using ROUGE metrics (ROUGE-1, ROUGE-2, ROUGE-L).
- Project source: [GitHub Link]

Dkash – Online Transaction Platform | *MongoDB, Express.js, React, Node.js* Fall 2023

- Collaborated in a team to design and develop a secure digital transaction platform using the MERN stack.
- Implemented software engineering practices, including requirements analysis, use-case modeling, system architecture, and interface design.
- Project source: [GitHub Link]

Technical Skills

Languages: Python, C/C++, JavaScript, Solidity
Frameworks & Libraries: React, Node.js, pandas, NumPy, Matplotlib, Pytorch
Developer Tools: VS Code, PyCharm, Code::Blocks, LTspice, emu8086
Core Competencies: Problem-solving, Team Collaboration, System Design

Languages

- **English:** Professional working proficiency
- **Bangla:** Native proficiency

References

Dr. Md Sadek Ferdous Professor
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Dr. S M Taiabul Haque Professor
Department of Computer Science and Engineering, BRAC University *Email: taiabul.haque@bracu.ac.bd*